

CHE 201 Spring 2020 HW 7

Thanh Nguyen

TOTAL POINTS

97 / 100

QUESTION 1

11 15 / 15

- **2 pts** Wrong schematic of the system
- **2 pts** Wrong engineering model
- **4 pts** wrong mass rate and energy rate balance
- **4 pts** Wrong exit temperature
- **3 pts** Wrong P-v diagram
- ✓ - **0 pts** Correct

QUESTION 2

2 2 15 / 15

- ✓ - **0 pts** Correct
- **2 pts** Wrong schematic of the system
- **2 pts** Wrong engineering model
- **4 pts** Wrong mass rate and energy rate balance
- **2 pts** Wrong mass flow rate of the air
- **2 pts** Wrong exit area
- **3 pts** Wrong T-v diagram

QUESTION 3

3 3 12 / 15

- **0 pts** Correct
- **2 pts** Wrong schematic of the system
- **2 pts** Wrong engineering model
- **4 pts** Wrong mass rate and energy rate balance
- **4 pts** Wrong power developed by the turbine
- ✓ - **3 pts** Wrong diagram P-V
- **1 pts** Incomplete engineering model

QUESTION 4

4 4 15 / 15

- ✓ - **0 pts** Correct
- **2 pts** Wrong schematic of the system
- **2 pts** Wrong engineering model
- **1 pts** Incomplete engineering model

- **4 pts** Wrong mass rate and energy rate balance
- **4 pts** Wrong heat transfer rate
- **3 pts** Wrong T-v diagram

QUESTION 5

5 5 20 / 20

- ✓ - **0 pts** Correct
- **4 pts** Wrong schematic of the system
- **4 pts** Wrong engineering model
- **6 pts** Wrong mass rate and energy rate balance
- **6 pts** Wrong the power required by the pump
- **1 pts** $W < 0$ work done on system

QUESTION 6

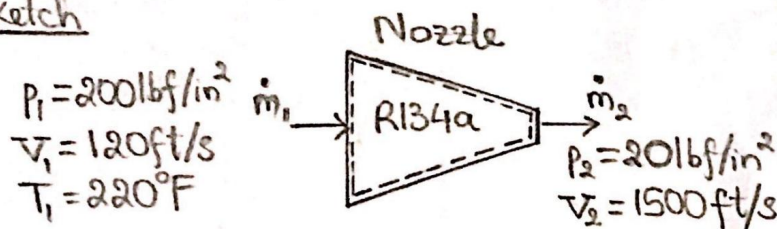
6 6 20 / 20

- ✓ - **0 pts** Correct
- **3 pts** Wrong schematic of the system
- **3 pts** Wrong engineering model
- **5 pts** Wrong mass rate and energy rate balance
- **5 pts** Wrong quality
- **4 pts** Wrong T-v diagram
- **2 pts** Incomplete engineering model

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1.)

(a) Sketch



(b) Engineering Model

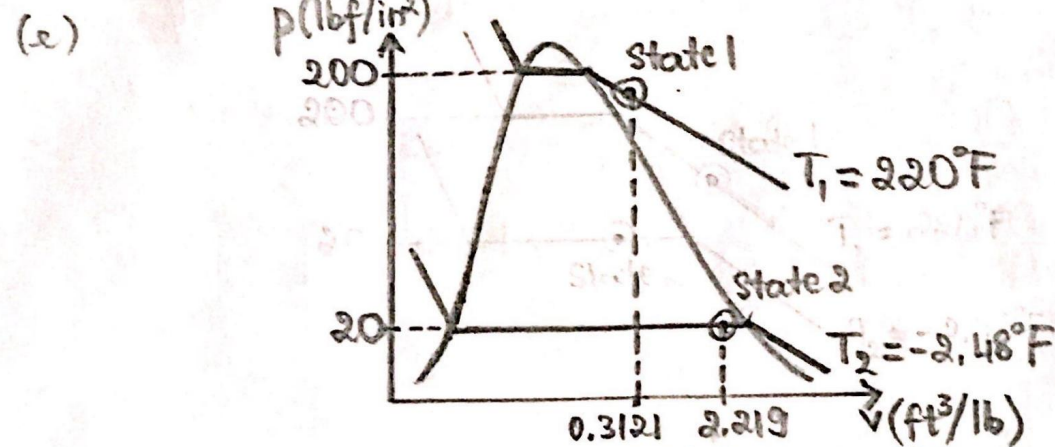
1. The system is a control volume at a steady state with 1 inlet and 1 outlet
2. $\dot{Q}_{cv} = \dot{W}_{cv} = \Delta PE = 0$

(c) Mass rate balance: $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_1 - \dot{m}_2 = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$
 Energy rate balance: $\frac{dE_{cv}}{dt} = 0 \rightarrow \underbrace{\dot{Q}_{cv}}_0 - \underbrace{\dot{W}_{cv}}_0 + \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g(z_1 - z_2) \right] = 0$
 $\Rightarrow \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} \right] = 0 \rightarrow (h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} = 0$

(d) Using table A-12E, at $p_1 = 200 \text{ lbf/in}^2$, $T_1 = 220^\circ\text{F} \rightarrow h_1 = 144.15 \text{ Btu/lb}$
 $\rightarrow h_2 = h_1 + \frac{1}{2}(V_1^2 - V_2^2) = 144.15 \text{ Btu/lb} + \frac{1}{2}[(120)^2 - (1500)^2] \left(\frac{\text{ft}}{\text{s}} \right)^2 \times \frac{1 \text{ lbf}}{32.174 \text{ lbf} \cdot \text{s}^2} \times \frac{1 \text{ Btu}}{778.2 \text{ lbf} \cdot \text{ft}}$
 $= 99.506 \text{ Btu/lb}$

Using table A-11E, at $p_2 = 20 \text{ lbf/in}^2$, $T = -2.48^\circ\text{F} \rightarrow h_f = 10.89 \text{ Btu/lb}$ and $h_g = 101.39 \text{ Btu/lb}$

Since $h_f < h_2 < h_g$ at $p_2 = 20 \text{ lbf/in}^2$, R134a at the exit is in a liquid-vapor mixture and $T_2 = -2.48^\circ\text{F}$



Using table A-12E, at $p_1 = 200 \text{ lbf/in}^2$, $T_1 = 220^\circ\text{F} \rightarrow v_1 = 0.3121 \text{ ft}^3/\text{lb}$

$$h_2 = (1-x)h_f + xh_g \rightarrow x = \frac{h_2 - h_f}{h_g - h_f} = \frac{99.506 - 10.89}{101.39 - 10.89} = 0.979$$

Using table A-11E, at $p_2 = 20 \text{ lbf/in}^2$, $T_2 = -2.48^\circ\text{F} \rightarrow v_f = 0.01181 \text{ ft}^3/\text{lb}$ and $v_g = 2.2661 \text{ ft}^3/\text{lb}$

$$v_2 = (1-x)v_f + xv_g = (1-0.979)(0.01181 \text{ ft}^3/\text{lb}) + 0.979(2.2661 \text{ ft}^3/\text{lb})$$

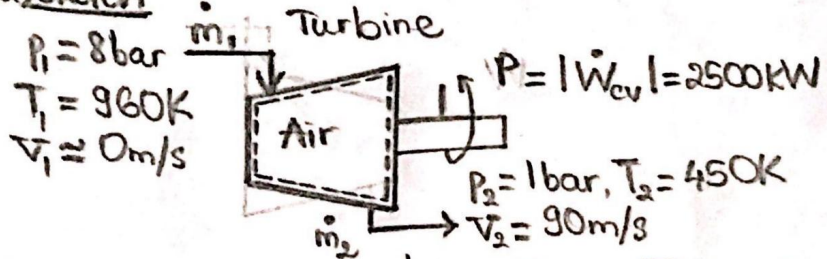
$$= 2.219 \text{ ft}^3/\text{lb}$$

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- 2 pts Wrong schematic of the system
- 2 pts Wrong engineering model
- 4 pts wrong mass rate and energy rate balance
- 4 pts Wrong exit temperature
- 3 pts Wrong P-v diagram
- ✓ - 0 pts Correct

2.)

(a) Sketch



(b) Engineering Model

- 1. The system is a control volume at a steady state with inlet and outlet
- 2. Ideal gas behavior
- 3. $V_1 \approx 0 \text{ m/s}$
- 4. $\dot{Q} = \Delta PE = 0$

(c) Mass rate balance: $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_1 - \dot{m}_2 = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$

Energy rate balance: $\frac{dE_{cv}}{dt} = 0 \rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g(z_1 - z_2)] = 0$
 $\rightarrow \dot{m}[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2}] = \dot{W}_{cv} \rightarrow \dot{m}[(h_1 - h_2) - \frac{V_2^2}{2}] = \dot{W}_{cv}$

(d) Using table A-22, at $T_1 = 960 \text{ K} \rightarrow h_1 = 1000.55 \text{ kJ/kg}$
at $T_2 = 450 \text{ K} \rightarrow h_2 = 451.80 \text{ kJ/kg}$

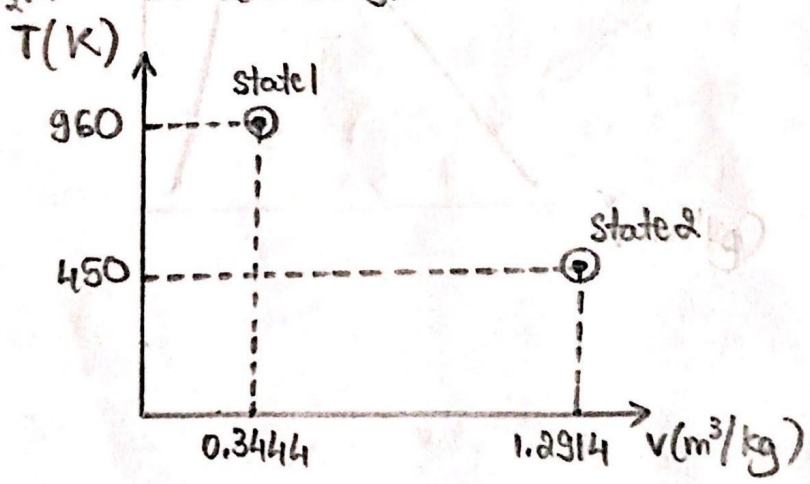
$$\rightarrow \dot{m} = \frac{\dot{W}_{cv}}{h_1 - h_2 - \frac{V_2^2}{2}} = \frac{2500 \text{ kW}}{1000.55 \text{ kJ/kg} - 451.80 \text{ kJ/kg} - \frac{1}{2}(90 \text{ m/s})^2 \times \frac{1 \text{ kJ}}{10^3 \text{ Nm}} \times \frac{1 \text{ N}}{1 \text{ kgm/s}^2}} \times \frac{1 \text{ kJ/s}}{1 \text{ kW}}$$

$$= 4.59 \text{ kg/s}$$
$$P_2 \bar{v}_2 = \bar{R} T_2 \rightarrow P_2 \bar{v}_2 M = \bar{R} T_2 \rightarrow \bar{v}_2 = \frac{\bar{R} T_2}{P_2 M}$$

$$\dot{m} = \frac{A_2 \bar{v}_2}{\bar{v}_2} \rightarrow A_2 = \frac{\dot{m} \bar{v}_2}{\bar{v}_2} = \frac{\dot{m} \bar{R} T_2}{\bar{v}_2 P_2 M} = \frac{(4.59 \text{ kg/s})(8.314 \text{ kJ/kmol} \cdot \text{K})(450 \text{ K})}{(90 \text{ m/s})(1 \text{ bar})(28.97 \text{ kg/kmol})} \times \frac{1 \text{ bar}}{10^5 \text{ N/m}^2} \times \frac{10^3 \text{ Nm}}{1 \text{ kJ}}$$
$$= 0.0659 \text{ m}^2$$

(e) $v_1 = \frac{\bar{R} T_1}{P_1 M} = \frac{(8.314 \text{ kJ/kmol} \cdot \text{K})(960 \text{ K})}{(8 \text{ bar})(28.97 \text{ kg/kmol})} \times \frac{1 \text{ bar}}{10^5 \text{ N/m}^2} \times \frac{10^3 \text{ Nm}}{1 \text{ kJ}} = 0.3444 \text{ m}^3/\text{kg}$

$$v_2 = \frac{\bar{R} T_2}{P_2 M} = \frac{(8.314 \text{ kJ/kmol} \cdot \text{K})(450 \text{ K})}{(1 \text{ bar})(28.97 \text{ kg/kmol})} \times \frac{1 \text{ bar}}{10^5 \text{ N/m}^2} \times \frac{10^3 \text{ Nm}}{1 \text{ kJ}} = 1.2914 \text{ m}^3/\text{kg}$$



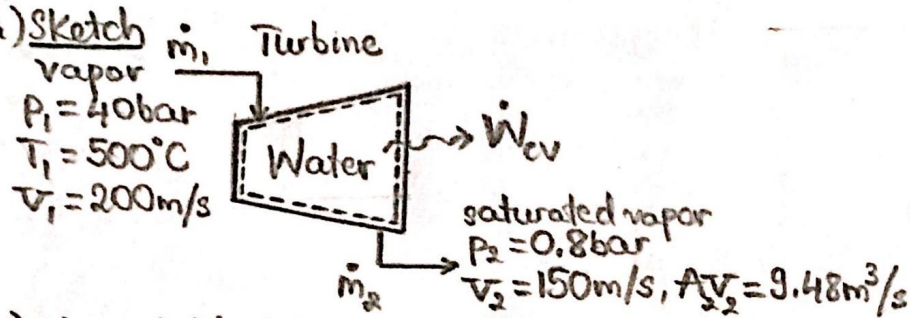
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✓ - 0 pts Correct

- 2 pts Wrong schematic of the system
- 2 pts Wrong engineering model
- 4 pts Wrong mass rate and energy rate balance
- 2 pts Wrong mass flow rate of the air
- 2 pts Wrong exit area
- 3 pts Wrong T-v diagram

3.)

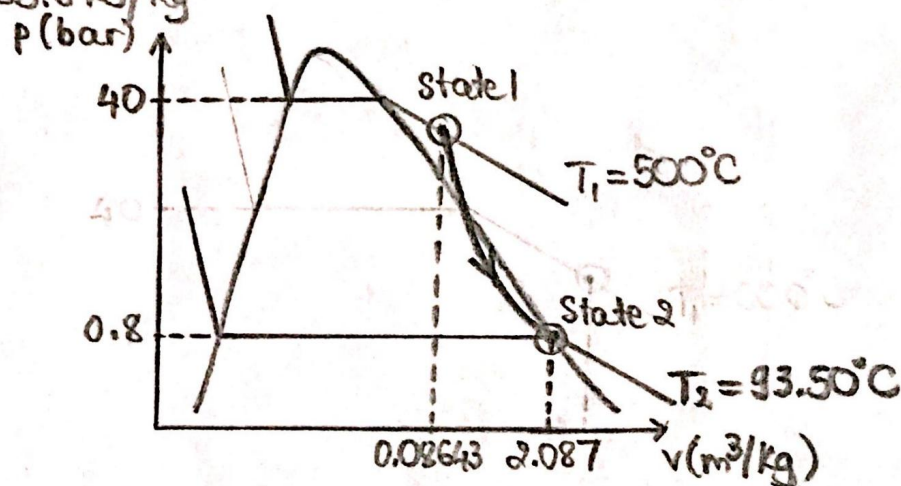
(a) Sketch



(c) Engineering Model

1. The system is a control volume at a steady state with inlet and outlet
2. Adiabatic process
3. $\Delta PE = Q = 0$

(b) Using table A-4, at $p_1 = 40 \text{ bar}$, $T_1 = 500^\circ\text{C} \rightarrow v_1 = 0.08643 \text{ m}^3/\text{kg}$, $h_1 = 3445.3 \text{ kJ/kg}$
 Using table A-3, saturated vapor at $p_2 = 0.8 \text{ bar} \rightarrow T_2 = 93.50^\circ\text{C}$, $v_2 = 2.087 \text{ m}^3/\text{kg}$,
 $h_2 = 2665.8 \text{ kJ/kg}$



(d) Mass rate balance: $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_1 - \dot{m}_2 = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$

Energy rate balance: $\frac{dE_{cv}}{dt} = 0 \rightarrow \cancel{\dot{Q}_{cv}} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \frac{v_1^2 - v_2^2}{2} + g(z_1 - z_2) \right] = 0$
 $\rightarrow \dot{m} \left[(h_1 - h_2) + \frac{v_1^2 - v_2^2}{2} \right] = \dot{W}_{cv}$

(e) $\dot{W}_{cv} = \dot{m} \left[(h_1 - h_2) + \frac{v_1^2 - v_2^2}{2} \right] = \frac{A_2 v_2}{v_2} \left[(h_1 - h_2) + \frac{v_1^2 - v_2^2}{2} \right]$

$$= \frac{9.48 \text{ m}^3/\text{s}}{2.087 \text{ m}^3/\text{kg}} \left[(3445.3 - 2665.8) \text{ kJ/kg} + \frac{(200^2 - 150^2)}{2} \left(\frac{\text{m}^2}{\text{s}^2} \right) \times \frac{1 \text{ kJ}}{10^3 \text{ Nm}} \times \frac{1 \text{ N}}{1 \text{ kg m/s}^2} \right]$$

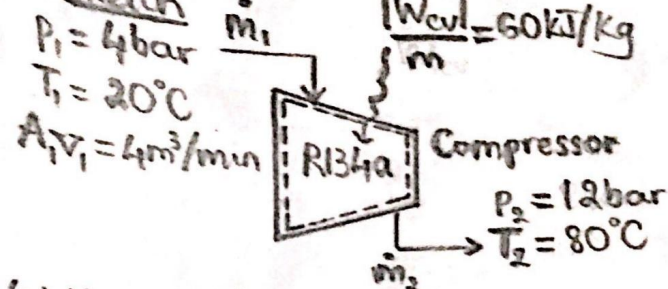
$$= 3580.6 \text{ kW}$$

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- 0 pts Correct
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- 4 pts Wrong mass rate and energy rate balance
- 4 pts Wrong power developed by the turbine
- ✓ - 3 pts Wrong diagram P-V
- 1 pts Incomplete engineering model

4.)

(a) Sketch

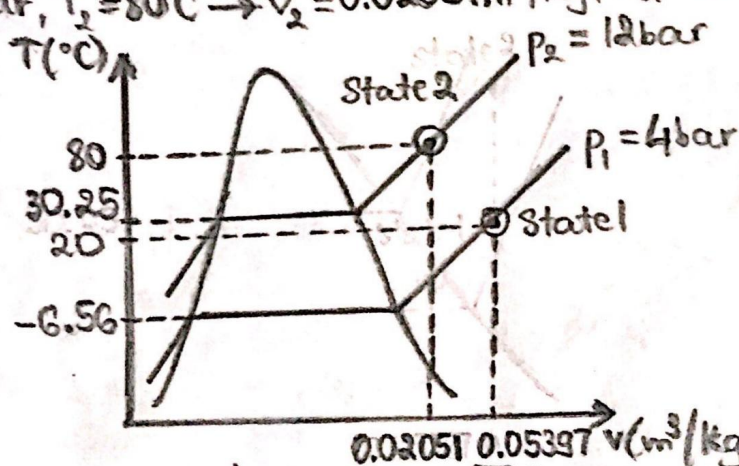


(c) Engineering Model

1. The system is a control volume at a steady state with inlet and outlet
2. $\Delta KE = \Delta PE = 0$

(b) Using table A-12

- $P_1 = 4 \text{ bar}, T_1 = 20^\circ\text{C} \rightarrow v_1 = 0.05397 \text{ m}^3/\text{kg}, h_1 = 262.96 \text{ kJ/kg}, T_{s1} = -6.56^\circ\text{C}$
- $P_2 = 12 \text{ bar}, T_2 = 80^\circ\text{C} \rightarrow v_2 = 0.02051 \text{ m}^3/\text{kg}, h_2 = 310.24 \text{ kJ/kg}, T_{s2} = 30.25^\circ\text{C}$



(d) Mass rate balance $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_1 - \dot{m}_2 = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$

Energy rate balance $\frac{dE_{cv}}{dt} = 0 \rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}[(h_1 - h_2) + \frac{v_1^2 - v_2^2}{2} + g(z_1 - z_2)] = 0$

$$\rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}(h_1 - h_2) = 0$$

$$(e) \dot{Q}_{cv} = \dot{W}_{cv} - \dot{m}(h_1 - h_2) = \left(\frac{\dot{W}_{cv}}{\dot{m}}\right) \dot{m} - \dot{m}(h_1 - h_2) = \frac{A_1 V_1}{v_1} \left(\frac{\dot{W}_{cv}}{\dot{m}} + h_2 - h_1\right)$$

$$= \frac{4 \text{ m}^3/\text{min}}{0.05397 \text{ m}^3/\text{kg}} \times \frac{1 \text{ min}}{60 \text{ s}} \times (-60 \text{ kJ/kg} + 310.24 \text{ kJ/kg} - 262.96 \text{ kJ/kg}) \times \frac{1 \text{ kW}}{1 \text{ kJ/s}}$$

$$= -15.71 \text{ kW}$$

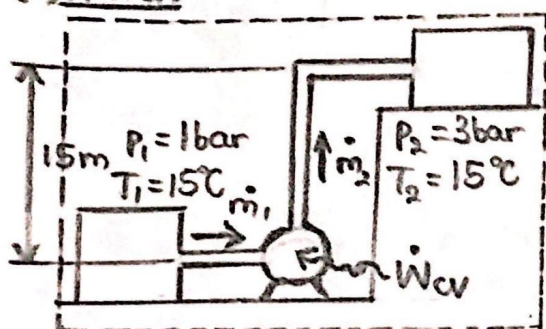
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✓ - 0 pts Correct

- 2 pts Wrong schematic of the system
- 2 pts Wrong engineering model
- 1 pts Incomplete engineering model
- 4 pts Wrong mass rate and energy rate balance
- 4 pts Wrong heat transfer rate
- 3 pts Wrong T-v diagram

5.)

(a) Sketch



(b) Engineering Model

1. The system is a control volume at a steady state with 1 inlet and 1 outlet

2. $g = 9.81 \text{ m/s}^2$

3. $Q = \Delta KE = 0$

4. $\Delta T \approx 0$

(c) Mass rate balance $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_1 - \dot{m}_2 = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m} = 1.5 \text{ kg/s}$

energy rate balance $\frac{dE_{cv}}{dt} = 0 \rightarrow \cancel{\dot{Q}_{cv}} - \dot{W}_{cv} + \dot{m}[(h_1 - h_2) + \cancel{\frac{V_1^2 - V_2^2}{2}} + g(z_1 - z_2)] = 0$

$\rightarrow \dot{W}_{cv} = \dot{m}[(h_1 - h_2) + g(z_1 - z_2)]$

(d) $h(T, p) \approx h_f(T) + v_f(T)[p - p_{sat}(T)]$

$T_1 = T_2 \rightarrow h_f(T_1) = h_f(T_2), p_{sat}(T_1) = p_{sat}(T_2)$

$\rightarrow h_1 - h_2 = v_f(T)(p_1 - p_2)$

Using table A-2, $v_f(15^\circ\text{C}) = 1.0009 \times 10^{-3} \text{ m}^3/\text{kg}$

$\rightarrow \dot{W}_{cv} = \dot{m}[v_f(15^\circ\text{C})(p_1 - p_2) + g(z_1 - z_2)]$

$= (1.5 \text{ kg/s})[(1.0009 \times 10^{-3} \text{ m}^3/\text{kg})(1 - 3) \text{ bar} \times \frac{10^5 \text{ N/m}^2}{1 \text{ bar}} + (9.81 \text{ m/s}^2)(-15 \text{ m})]$

$= -521 \text{ W} = -0.521 \text{ kW}$

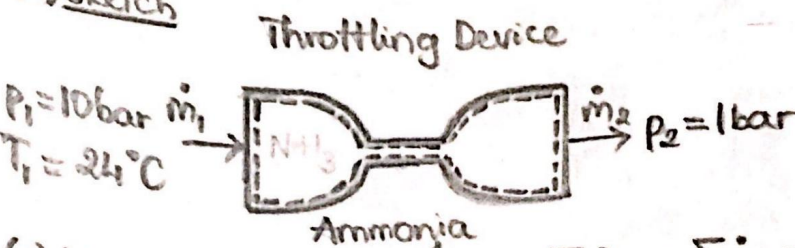
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✓ - 0 pts Correct

- 4 pts Wrong schematic of the system
- 4 pts Wrong engineering model
- 6 pts Wrong mass rate and energy rate balance
- 6 pts Wrong the power required by the pump
- 1 pts $W < 0$ work done on system

5.)

(a) Sketch



(b) Engineering Model

1. The system is a control volume at a steady state with 1 inlet and 1 outlet
2. $Q = W_{cv} = \Delta KE = \Delta PE = 0$

(c) Mass rate balance: $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_1 - \dot{m}_2 = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$

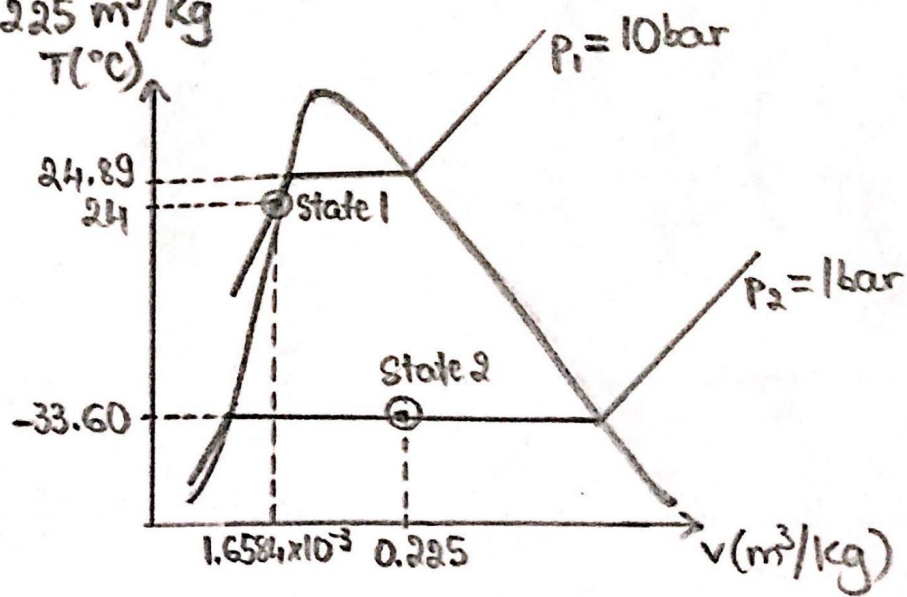
Energy rate balance: $\frac{dE_{cv}}{dt} = 0 \rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g(z_1 - z_2)] = 0$
 $\rightarrow \dot{m}(h_1 - h_2) = 0 \rightarrow h_1 = h_2$

(d) Using table A-14, at $p_1 = 10 \text{ bar} \rightarrow T_{sat} = 24.89^\circ\text{C}$, $v_f = 1.6584 \times 10^{-3} \text{ m}^3/\text{kg}$ and

$h_f = 297.76 \text{ kJ/kg}$
 $T_1 = 24^\circ\text{C} < T_{sat} \rightarrow \text{state 1 is a compressed liquid}$
 $\rightarrow h_1 \approx h_f = 297.76 \text{ kJ/kg}$, $v_1 \approx v_f = 1.6584 \times 10^{-3} \text{ m}^3/\text{kg}$
 $\rightarrow h_2 = h_1 = 297.76 \text{ kJ/kg}$

Using table A-14, at $p_2 = 1 \text{ bar} \rightarrow T_2 = -33.60^\circ\text{C}$
 $h_g = 1398.41 \text{ kJ/kg}$
 $v_g = 1.1381 \text{ m}^3/\text{kg}$
 $h_2 = (1-x)h_f + xh_g \rightarrow x = \frac{h_2 - h_f}{h_g - h_f} = \frac{297.76 - 28.18}{1398.41 - 28.18} = 0.1967$

(e) $v_2 = (1-x)v_f + xv_g = (1-0.1967)(1.6584 \times 10^{-3} \text{ m}^3/\text{kg}) + (0.1967)(1.1381 \text{ m}^3/\text{kg})$
 $= 0.225 \text{ m}^3/\text{kg}$



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✓ - 0 pts Correct

- 3 pts Wrong schematic of the system
- 3 pts Wrong engineering model
- 5 pts Wrong mass rate and energy rate balance
- 5 pts Wrong quality
- 4 pts Wrong T-v diagram
- 2 pts Incomplete engineering model